

"99.9% of science is boring."
- Makise Kurisu

PRECALCULUS

August 13, 2026

Wisam Yassar Mahardika

CONTENTS

1	Functions	1
1.1	Intro To Functions	1
1.2	Evaluating Functions	3
1.2.1	Difference Quotient	4
1.3	Finding Domains	5
1.3.1	The Problem Areas	5
1.4	Operations Of Functions	7
1.5	The Vertical Line Test	8
1.6	Features of Graphs Points, Domain, Range	9
1.7	Properties of Functions: Even or Odd	11
1.8	Properties of Functions: Increasing or Decreasing	13
1.9	Properties of Functions: Extremas	15
1.10	Average Rate of Change	16
1.11	Piece-wise Functions	17
1.12	Important Functions	19
2	Graphical Transformations	22
2.1	Introduction To Graphical Transformations	22
2.1.1	Vertical	22
2.1.2	Horizontal	22
2.1.3	Vertical Stretch/Compression	24
2.1.4	Horizontal Stretch/Compression	24
2.1.5	Reflection	25
2.2	How to Graph with Transformations	27
3	Quadratics	29
3.1	Intro to Solving Quadratics	29
3.1.1	Cases	29
3.2	The Square Root Method for Quadratics	31
3.3	Factoring Method for Quadratics	33
3.4	Completing the Square Method for Quadratics	36
3.5	Proving the Quadratic Formula	39
3.5.1	Proof	39
3.6	The Quadratic Formula	40
3.7	Graphs of Quadratics	42
3.8	Quadratics Inequalities	46

4	Intersection of Functions	48
5	Solving Equations by Using Substitutions	50
6	Polynomials	53
6.1	Introduction to Polynomials	53
6.2	Power Functions	55
6.3	Multiplicity and End Behavior of Polynomials	58
6.4	Polynomial functions from x-intercepts and graphs	59
6.5	How to Sketch Polynomial Functions	62
6.6	Synthetic Division and Long Division of Polynomials	66

$$\begin{array}{r}
 3x^5 - 9x^4 + 27x^3 + x^2 - 3x + 9 \\
 x+3 \overline{) \quad 3x^6 + 82x^3 + 27} \\
 \underline{-3x^6 - 9x^5} \\
 -9x^5 \\
 \underline{9x^5 + 27x^4} \\
 27x^4 + 82x^3 \\
 \underline{-27x^4 - 81x^3} \\
 x^3 \\
 \underline{-x^3 - 3x^2} \\
 -3x^2 \\
 \underline{3x^2 + 9x} \\
 9x + 27 \\
 \underline{-9x - 27} \\
 0
 \end{array}$$

$$g(x) = (x+3)(3x^5 - 9x^4 + 27x^3 + x^2 - 3x + 9)$$

2. Synthetic Division:

$$\begin{array}{r|rrrrrrrr}
 -3 & 3 & 0 & 0 & 82 & 0 & 0 & 27 \\
 & \downarrow & + & - & 9 & + & 27 & + & - & 81 & + & - & 3 & + & 9 & + & - & 27 \\
 \hline
 & 3 & -9 & 27 & 1 & -3 & 9 & 0
 \end{array}$$

$$g(x) = (x+3)(3x^5 - 9x^4 + 27x^3 + x^2 - 3x + 9)$$

Example 6.13 (Example with a Non Zero Remainder).

$$h(x) = 4x^3 - 3x^2 - 8x + 4; x = 2$$

$$h(2) = 8$$

1. Long Division:

$$\begin{array}{r}
 4x^2 + 5x + 2 \\
 x-2 \overline{) \quad 4x^3 - 3x^2 - 8x + 4} \\
 \underline{-4x^3 + 8x^2} \\
 5x^2 - 8x \\
 \underline{-5x^2 + 10x} \\
 2x + 4 \\
 \underline{-2x + 4} \\
 8
 \end{array}$$

$$4x^2 + 5x + 2 + \frac{8}{x-2}$$

$$h(x) = (x-2)(4x^2 + 5x + 2 + \frac{8}{x-2})$$

2. Synthetic Division:

$$\begin{array}{r|rrrr}
 2 & 4 & -3 & -8 & 4 \\
 & \downarrow & + & 8 & + & 10 & + & 4 \\
 \hline
 & 4 & 5 & 2 & 8
 \end{array}$$

$$4x^2 + 5x + 2 + \frac{8}{x-2}$$

$$h(x) = (x-2)(4x^2 + 5x + 2 + \frac{8}{x-2})$$